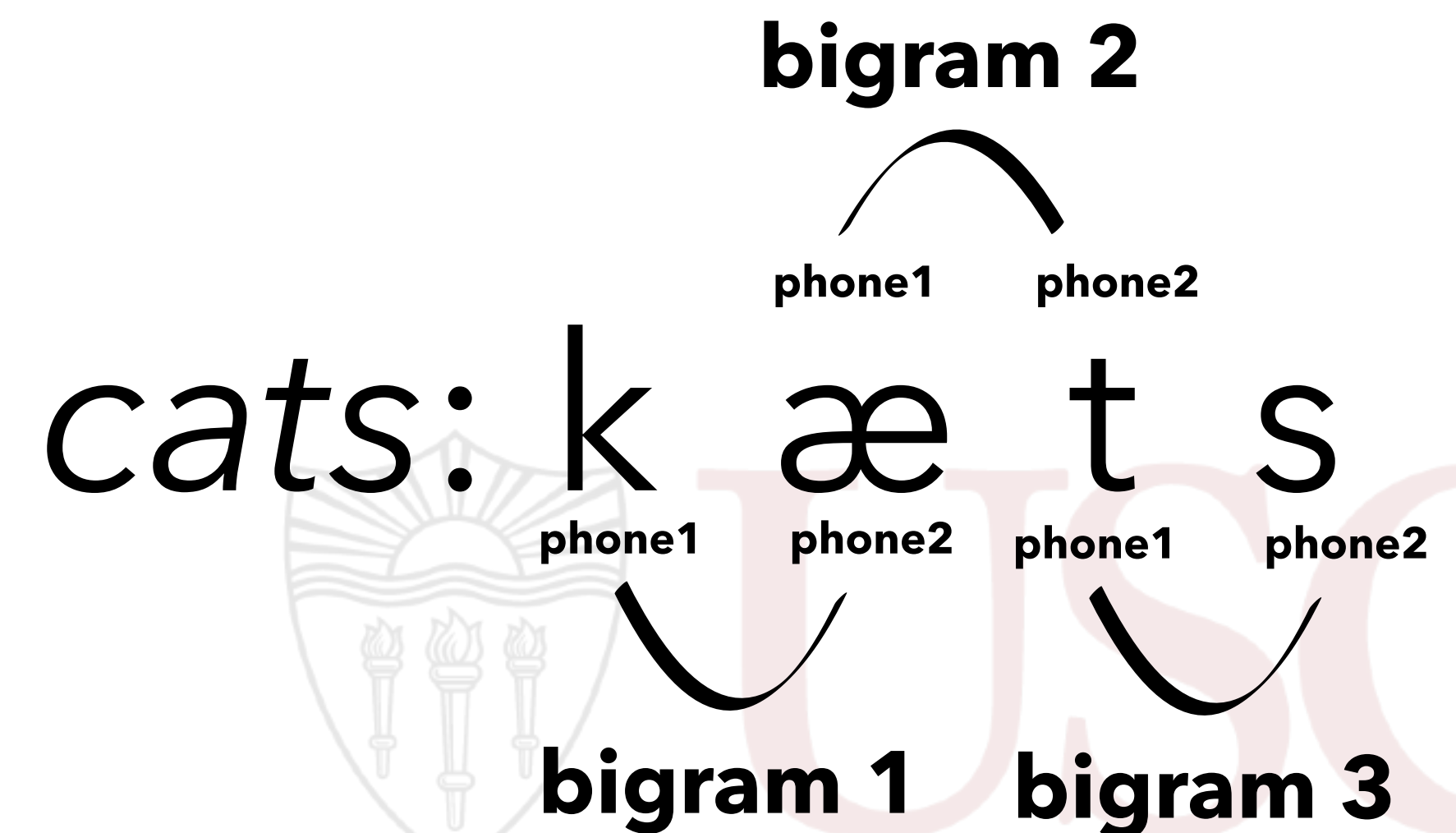




# General distinctiveness as an estimate of variation

- To investigate the effects of information-theoretic constraints on speech variation in an unsupervised way:
- This study focuses on the articulatory & acoustic “distinctiveness” (resilience to variation) between phone bigrams (phone1-phone2) in a word.





# General distinctiveness as an estimate of variation

- To investigate the effects of information-theoretic constraints on speech variation in an unsupervised way:
  - How does the contextual information content (**surprisal**) of the bigram, given its previous context in the word, affect the phone2's distinctiveness to phone1?

$$I(\text{bigram} \mid \text{context}) = -\log(P(\text{bigram} \mid \text{context}))$$

$$I(\mathbf{k}\mathbf{\ae} \mid \# \_\_) = -\log(P(\mathbf{k}\mathbf{\ae} \mid \# \_\_))$$

$$I(\mathbf{\ae}\mathbf{t} \mid \mathbf{k} \_\_) = -\log(P(\mathbf{\ae}\mathbf{t} \mid \mathbf{k} \_\_))$$

$$I(\mathbf{ts} \mid \mathbf{k}\mathbf{\ae} \_\_) = -\log(P(\mathbf{ts} \mid \mathbf{k}\mathbf{\ae} \_\_))$$

